
USER GUIDE

GlucoTrack One

Digital Blood Glucose Monitoring System

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The GlucoTrack One is an in vitro diagnostic (IVD) medical device intended for self-monitoring of blood glucose levels. The information contained in this document is believed to be accurate at the time of publication; however, no warranties, express or implied, are provided regarding its completeness or accuracy.

Users are responsible for following all instructions provided in this guide. The manufacturer shall not be liable for any damage resulting from improper use of the device or failure to adhere to the instructions.

Always consult a qualified healthcare professional for medical advice and interpretation of results.

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1. Safety Information

Read Before Use: This guide provides essential instructions for safe operation of the GlucoTrack One device.



WARNINGS

These warnings indicate conditions that may result in injury or inaccurate readings.



Do not use on infants or individuals unable to remain still.



Consult a healthcare professional before interpreting results.



Do not use the device if it is damaged or malfunctioning.



Keep the device and accessories out of reach of children.



Do not expose the device to extreme heat, cold, or direct sunlight.



Use only the specified meter and components for this device system.



CAUTIONS

These cautions indicate conditions that may result in minor injury or reading errors.



Always test on clean, dry hands. Hand contamination affects accuracy.



Ensure test strip is fully and correctly inserted. Do not reuse old or bend strips.



Keep the device dry. Wipe with a clean, dry cloth. Do not use water or solvents.



NOTES

These notes provide important usage tips and reminders for optimal device operation.



ROTATE LANCE SITES

Alternate fingers and sites (e.g., sides of tips) to avoid calluses and maintain sample consistency.



SEPARATE STORAGE

Store test strips and the meter in their case. Discard old batteries properly and keep them away from the meter.



CLEAN PORT GENTLY

Periodically use a clean, dry, soft brush to remove dust and debris from the test strip port. Do not insert foreign objects.

2. Introduction

This section provides an overview of the GlucoTrack One Digital Blood Glucose Monitoring System, including its purpose, features, and basic functionality.

GlucoTrack One is a portable device designed to measure blood glucose levels quickly and accurately using a small blood sample. It uses electrochemical sensing technology to analyze glucose concentration and display results on a digital screen.

2.1. Product Overview

The GlucoTrack One is a compact, battery-operated device suitable for both home and clinical use. It provides fast reading and supports memory storage for tracking previous results.

- Display screen for showing test results and indicators
- Test strip port for inserting disposable strips
- Control button for navigation and operation
- Battery compartment for power supply
- Internal memory module for storing readings.

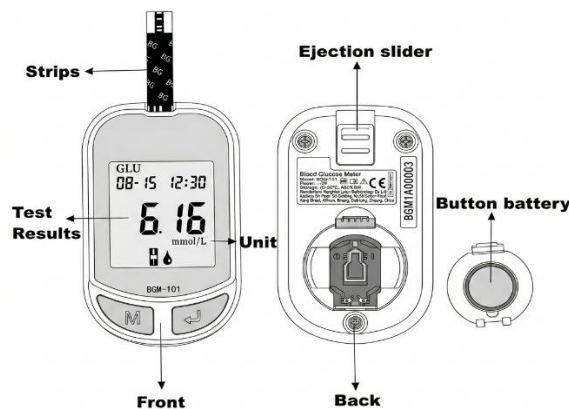


Figure 1. Product Overview

2.2. Key Features

The GlucoTrack One includes the following features:

- Rapid measurement with results displayed within seconds
- Digital LCD screen for clear and readable output
- Memory function for storing previous readings
- Simple one-button operation for ease of use
- Compact and lightweight design for portability
- Automatic power-off to conserve battery life.

2.3. Intended Use

The GlucoTrack One is intended for the quantitative measurement of glucose levels in fresh capillary whole blood samples.

The device is suitable for:

- Self-monitoring by individuals at home
- Use by healthcare professionals in clinical settings.

2.4. Environmental Conditions for Use

For accurate measurement results, the device should be operated under the following conditions:

- Temperature range: 10°C to 40°C
- Humidity: Within recommended limits
- Clean and dry environment.

Operating the device outside these conditions may affect accuracy.

3. Package Contents

This section lists all components included with the GlucoTrack One Digital Blood Glucose Monitoring System. Each component is designed to support accurate measurement and proper operation of the device.

3.1. Included Items

Table 1. Package Content

Item	Component	Description	Key Specification
1.	Glucose Meter	Main device used to measure blood glucose levels	Digital display, battery-operated
2.	Test Strips	Disposable strips used for glucose measurement	Single-use, electrochemical sensing
3.	Lancet Device	Device used to obtain a small blood sample	Adjustable depth settings
4.	Lancets	Sterile needles used with lancet device	Single-use, sterile
5.	Battery	Power source for the device	2 × AAA (1.5V)
6.	Carrying Case	Storage case for devices and accessories	Portable, protective
7.	User Guide	Detailed instructions for operation and maintenance	Printed manual
8.	Quick Start Guide	Basic setup and usage instructions	Reference guide



Figure 2. Included Items

3.2. Component Description

Each component included in the package serves a specific function:

- The glucose meter processes and displays measurement results
- Test strips are used to collect and analyze blood samples
- The lancet device and lancets are used to safely obtain a blood sample
- The battery powers the device for operation
- The carrying case protects the device during storage and transport.

3.3. Package Verification

Before initial use:

- Ensure all components listed in [Table 1](#) are included
- Check for visible damage or missing items
- Confirm that test strips are within their expiration date.

4. Symbols and Markings

This section describes the symbols and indicators that may appear on the device display. These symbols provide important information about device status, measurement conditions, and alerts.

Understanding these symbols helps ensure correct operation and accurate interpretation of results.

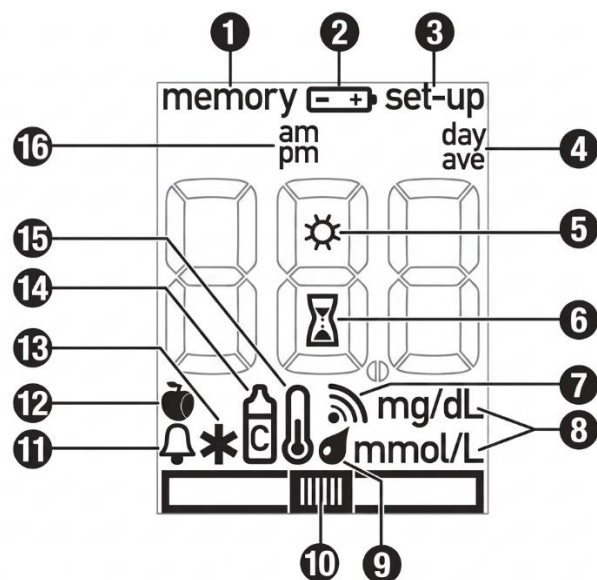


Figure 3. Symbols and Markings

4.1. Explanation of Symbols

No.	Symbol	Meaning
1.	memory	You are in the test-result memory. Either a saved test result or an average is being displayed.
2.		The battery is almost empty. When the symbol first appears, you can perform approx. 50 more tests.
3.	set-up	You are in the settings for the time format, time, date and beep.
4.	day ave	When an average is displayed (memory): the symbol is preceded by the number of days taken into account.
5.		Together with error message E-5: The meter is exposed to too much light.
6.		Flashing – A test is in progress or an average is being calculated.
7.		The beep is turned on.
8.	mg/dL or mmol/L	The unit your meter uses to display test results, either mg/dL or mmol/L depending on the meter.
9.		Flashing blood drop symbol – Now you can apply blood or control solution to the test strip.
10.		Flashing blood drop and test strip symbol. The test strip was removed from the meter before blood or control solution was applied, for example to apply blood outside the meter.
11.		Test reminder together with the symbol for a blood glucose test after a meal.
12.		“Before meal” or “After meal” marker
13.		“Other” marker

14.		Marker for control tests
15.		At the time of testing, the temperature is or was between +5 and +8 °C or between +42 and +45 °C
16.		Additional time information if 12-hour time format is set.

5. Operating Instructions

This section provides detailed instructions for preparing, operating and maintaining the GlucoTrack One device during routine use.

Accurate results depend on proper handling, preparation and adherence to the steps outlined below.

5.1. Preparing the Device for Use

Before performing a measurement, ensure that the device is properly set up and ready for operation.

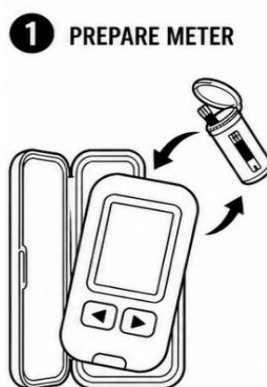
1. Open the battery compartment located at the back of the device.
2. Insert the batteries according to the polarity markings (+/-).
3. Close the battery compartment securely.
4. Ensure the device powers on correctly.
5. Verify that the display screen is functioning without errors.

5.2. Performing a Blood Glucose Test

This section describes the correct procedure for obtaining and measuring a blood sample.

Step 1: Prepare the Meter

Open the device and ensure that it is powered on and ready for operation. Verify that the display is functioning correctly and no error messages are shown.



Step 2: Insert Test Strip

Insert a new test strip into the test strip port. The device will automatically detect the strip and enter measurement mode.

Note: Ensure that the test strip is properly inserted to avoid errors.

2 INSERT STRIP

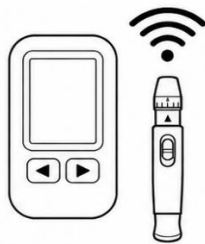


Step 3: Gather Supplies

Ensure that all required materials are available:

- Test strip
- Lancet device
- Lancet
- Cleaning material (if required)

3 GATHER SUPPLIES



Step 4: Clean the Finger

Clean the fingertip using an appropriate method (e.g., alcohol swab) and allow it to dry completely before sampling.

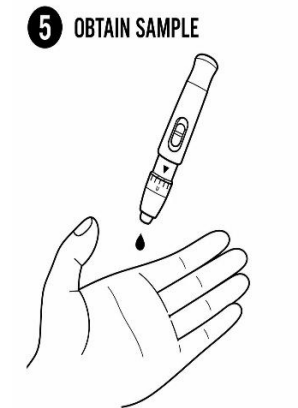
Caution: Moisture may affect the accuracy of the result.

4 CLEAN FINGER



Step 5: Obtain Blood Sample

Use the lancet device to obtain a small blood sample from the fingertip. Ensure that an adequate sample is collected.



Step 6: Read the Result

Apply the blood sample to the test strip. The device will process the sample and display the glucose value within a few seconds.



Step 7: Record the Result

Review the displayed result and record it if required. The device may also store the reading automatically in memory.

7 RECORD RESULT



Step 8: Dispose and Complete

Remove and safely dispose of the used test strip and lancet according to medical waste guidelines. Store the device properly after use.

Warning: Used components are biohazardous and must be handled appropriately.

8 DISPOSE & RECORD



5.3. Understanding the Measurement Result

After the test is completed, the result is displayed on the screen along with relevant indicators. The display may include:

- Glucose value
- Measurement unit (mg/dL or mmol/L)
- Time and date
- Additional symbols (if applicable)

5.4. Storing and Reviewing Results

The device automatically stores measurement results for future reference. Procedure to view stored results:

1. Activate memory mode using the control button.
2. Scroll through stored readings.
3. Review historical values and timestamps.

5.5. Completing the Test

After the measurement is complete, the used components must be handled properly.

1. Remove the test strip from the device.
2. Dispose of the test strip according to local medical waste guidelines.
3. Safely discard the lancet after use.

5.6. Power Management

The device includes automatic and manual power control features.

- The device automatically powers off after a period of inactivity
- Manual power-off can be performed using the control button. This helps conserve battery life and maintain device readiness.

5.7. Storage After Use

Proper storage is essential to maintain device performance and longevity.

- Store the device in a clean and dry environment
- Avoid exposure to extreme temperatures
- Keep the device in the provided carrying case when not in use

Note: Proper storage helps maintain measurement accuracy over time.

6. Troubleshooting

This section provides guidance for identifying and resolving common issues that may occur during the operation of the GlucoTrack One device. If a problem persists after performing the recommended actions, discontinue use and contact customer support.

6.1. General Troubleshooting Guidelines

Before performing specific troubleshooting steps:

- Ensure the device is powered on correctly
- Verify that the test strip is properly inserted
- Confirm that all components are clean and undamaged
- Check that the device is operated within the recommended environmental conditions.

6.2. Common Issues and Solutions

Table 2. Troubleshooting Guide

Issue	Possible Cause	Recommended Action
Device does not power on	Battery depleted or inserted incorrectly	Replace batteries and check polarity
No display or blank screen	Device malfunction or low battery	Restart device or replace batteries
Test strip not detected	Improper insertion or damaged strip	Reinsert strip or use a new one
Inaccurate reading	Insufficient blood sample or improper testing conditions	Repeat test with correct procedure
Device shuts off unexpectedly	Low battery or inactivity timeout	Replace battery or restart device
Error message displayed	Device or measurement error	Refer to error codes section

6.3. Error Codes and Indicators

Table 3. Error Codes

Error Code	Description	Action
E1	Test strip error	Insert a new test strip correctly
E2	Insufficient sample	Apply adequate blood sample and retest
E3	Measurement failure	Repeat the test with a new strip
E4	Temperature out of range	Move device to recommended environment
E5	Excessive light exposure	Perform test in a shaded or indoor environment

6.4. Environmental and Handling Issues

Certain conditions may affect device performance:

- Exposure to extreme temperatures
- High humidity or moisture
- Direct sunlight during testing
- Use of expired or contaminated test strips.

Caution: Operating the device outside recommended conditions may result in inaccurate reading.

6.5. Preventive Measures

To minimize operational issues:

- Use only approved test strips and accessories
- Store the device properly after use
- Regularly clean and maintain the device
- Follow all operating instructions carefully.

7. Maintenance and Care

This section provides instructions for maintaining the GlucoTrack One device to ensure accurate measurements, safe operation and long-term reliability. Proper maintenance reduces the risk of errors and extends the lifespan of the device.

7.1. General Maintenance Guidelines

To maintain optimal device performance:

- Keep the device clean and free from dust and contaminants
- Handle the device with care to avoid physical damage
- Avoid dropping or exposing the device to strong impact
- Use only approved accessories and test strips.

7.2. Cleaning the Device

The device should be cleaned regularly to maintain hygiene and performance.

- Turn off the device before cleaning
- Use a soft, dry cloth to wipe the exterior surfaces

- If required, lightly dampen the cloth with a mild cleaning solution
- Ensure the device is completely dry before use.

Caution: Moisture entering the device may cause malfunction or damage.



Figure 4. Cleaning the device

7.3. Handling Test Strips and Lancets

Proper handling of consumables is essential for accurate results.

- Store test strips in their original container
- Keep the container tightly closed after use
- Do not use expired or damaged test strips
- Use each lancet only once.

Warning: Reuse of lancets or test strips may lead to incorrect readings or infection risk.

7.4. Battery Maintenance

Proper battery handling ensures uninterrupted device operation.

- Replace batteries when the low battery indicator appears
- Use only specified battery type (e.g., AAA 1.5V)
- Remove batteries if the device will not be used for an extended period.

Note: Battery leakage may damage the device if left unused for long durations.

7.5. Disposal Guidelines

Dispose of device components responsibly:

- Used test strips and lancets must be treated as medical waste
- Dispose of batteries according to local regulations
- Do not discard electronic components with general waste.

Caution: Improper disposal may pose health or environmental risks.

8. Technical Specifications

This section provides the technical details of the GlucoTrack One Digital Blood Glucose Monitoring System. These specifications define the operational limits, performance characteristics, and environmental requirements of the device.

8.1. Measurement Specifications

Table 4. Measurement Specifications

Parameter	Specification
Measurement Range	20 – 600 mg/dL (1.1 – 33.3 mmol/L)
Measurement Principle	Electrochemical biosensor
Sample Type	Fresh capillary whole blood
Sample Volume	Approximately 0.5 μ L
Test Time	Approximately 5–10 seconds
Units of Measurement	mg/dL or mmol/L
Measurement Accuracy	Within $\pm 5\%$ under standard conditions
Calibration	Factory calibrated

Note: Measurement accuracy may vary depending on environmental conditions and sample handling.

8.2. Device and Operating Conditions

Table 5. Device and Operating Conditions

Parameter	Specification
Power Source	2 $\times$ AAA (1.5 V) batteries
Battery Life	Approximately 500–1000 tests
Memory Capacity	Stores up to 100–300 readings
Data Storage	Date, time, and results
Average Calculation	7, 14, and 30-day averages
Operating Temperature	10°C to 40°C
Storage Temperature	5°C to 45°C
Operating Environment	Clean and dry conditions

Caution: Operation outside recommended environmental conditions may affect accuracy and device performance.

8.3. Physical Characteristics and Compliance

Table 6. Physical Characteristics and Compliance

Parameter	Specification
Device Type	Portable handheld meter
Display	Digital LCD
Design	Compact and lightweight
Usage	Home and clinical environments
Device Classification	Medical device (self-monitoring)
Compliance	Meets applicable safety standards

9. Warranty and Customer Support

This section provides information about the warranty coverage and support services available for the GlucoTrack One Digital Blood Glucose Monitoring System.

9.1 Warranty Coverage

The GlucoTrack One device is covered by a limited warranty against manufacturing defects under normal use conditions.

Warranty Terms

- The warranty period is 12 months from the date of purchase
- The warranty applies to the main device unit only
- Repairs or replacements are provided at no additional cost during the warranty period.

Warranty Limitations

- Damage caused by misuse, improper handling or unauthorized modifications
- Damage resulting from accidents, drops or exposure to liquids
- Use of non-approved accessories or test strips
- Normal wear and tear of consumable components (e.g., test strips, lancets, batteries).

Note: Proof of purchase may be required to validate warranty claims.

9.2 Service and Support

For technical assistance or service-related inquiries, contact the authorized support center.

Support Services Include

- Troubleshooting assistance
- Device inspection and repair
- Replacement of defective units (within warranty terms)
- General product usage guidance.

When to Contact Support

- Persistent error messages
- Device malfunction or physical damage
- Inconsistent or unreliable readings
- Device fails to power on after battery replacement.

9.3 Customer Responsibilities

To ensure proper warranty coverage and device performance:

- Follow all instructions provided in this user guide
- Use only approved accessories and consumables
- Maintain the device according to recommended guidelines
- Store the device under appropriate conditions.

10. Regulatory and Compliance Information

This section provides information regarding the regulatory classification, safety standards, and compliance requirements applicable to the GlucoTrack One Digital Blood Glucose Monitoring System.

10.1 Medical Device Classification

The GlucoTrack One is classified as a medical device for self-monitoring of blood glucose levels. It is intended for use in both home and clinical environments under standard operating conditions.

The classification varies across regulatory regions as shown below:

Table 7. Medical Device Classification by Region

Region / Country	Regulatory Authority	Classification	Regulatory Framework / Requirement
United States	FDA (Food and Drug Administration)	Class II	Requires 510(k) premarket notification
European Union	Notified Bodies (EU MDR / IVDR)	Class IIa / Class C (IVD)	Requires CE marking and conformity assessment
India	CDSCO (Central Drugs Standard Control Organization)	Class B	Requires registration under Medical Device Rules
United Kingdom	MHRA (Medicines and Healthcare products Regulatory Agency)	Class IIa	Requires UKCA marking
Australia	TGA (Therapeutic Goods Administration)	Class IIa	Must be listed in ARTG
Canada	Health Canada	Class II	Requires Medical Device License (MDL)

10.2 Compliance Standards

The device is manufactured and tested in accordance with applicable international standards and guidelines, including:

- General safety requirements for medical electrical equipment
- Performance standards for blood glucose monitoring systems
- Quality management system standards for medical device manufacturing.

10.3 Safety and Usage Compliance

To maintain compliance and ensure safe operation:

- Use the device only as described in this user guide
- Operate the device within specified environmental conditions
- Use only approved accessories and consumables
- Follow proper cleaning, maintenance, and disposal procedures

Caution: Failure to follow these instructions may affect device performance and safety compliance.

10.4 Manufacturer Disclaimer

The GlucoTrack One device is intended for monitoring purposes only and should not be used as the sole basis for diagnosis or medical decisions.

Appendix 1: Glossary of Terms

This glossary provides definitions of key terms used throughout this user guide. Understanding these terms helps ensure correct operation of the device and accurate interpretation of results.

Table 8. Glossary of Terms

Term	Definition
Blood Glucose	The concentration of glucose (sugar) present in the blood, typically measured to monitor metabolic health.
Calibration	The process of ensuring that the device provides accurate measurement results. The device is factory calibrated and does not require manual adjustment.
Capillary Blood	Blood obtained from a fingertip sample, commonly used for glucose testing.
Control Solution	A liquid solution with a known glucose concentration used to verify that the device and test strips are functioning correctly.
Electrochemical Measurement	A method used by the device to measure glucose levels by detecting electrical signals generated from a chemical reaction on the test strip.
Error Code	A code displayed by the device indicating a specific issue or condition that requires attention.
Glucose Meter	A medical device used to measure the concentration of glucose in blood samples.
Lancet	A small, sterile needle used to obtain a blood sample from the fingertip.
Lancet Device	A spring-loaded device used to safely and consistently obtain a blood sample using a lancet.
Measurement Unit (mg/dL or mmol/L)	Units used to display blood glucose levels. mg/dL is commonly used in the United States, while mmol/L is used in many other regions.
Memory Function	A feature that allows the device to store previous test results along with date and time for review.
Test Strip	A disposable component used with the device to collect and analyze a blood sample for glucose measurement.
Self-Monitoring	The process of regularly checking blood glucose levels using a personal device.
IVD (In Vitro Diagnostic Device)	A medical device used to perform tests on samples such as blood outside the human body.
Hypoglycemia	A condition characterized by abnormally low blood glucose levels.
Hyperglycemia	A condition characterized by abnormally high blood glucose levels.
Calibration	The process of ensuring that the device provides accurate measurement results. The device is factory calibrated and does not require manual adjustment.